2024044

B.E. II EXAMINATION DEC.' 2023
Civil Engg.
3AVRC1: Applied Mathematics-III

Time: 3 Hrs.] **[Max. Marks:** 60

Note: Attempt any **TWO** parts from each Question. Each carries equal marks.

Q.1. (a) Evaluate: (06)

(i) $L\{t\cos at\}$ (ii) $L\left\{\int_0^t \frac{e^{-t}\sin t}{t}\,dt\right\}$

(b) Using convolution theorem, evaluate $\mathbf{L^{-1}}\left\{\frac{1}{(s^2+a^2)^2}\right\}$. (06)

(c) Solve by method of Laplace transform the equation $\mathbf{y''' + 2y'' - y' - 2y = 0}$, given that $\mathbf{y(0) = 0,\ y'(0) = 0, y''(0) = 6}$. (06)

Q.2. (a) 100 students of management institute obtained the following grade in statistics paper: (06)

Grade	A	B	C	D	E	Total
Frequency	15	17	30	22	16	100

Using χ^2 test, examine the hypothesis that the distribution of grades is uniform. Given that $\chi_4^2(0.05) = 9.49$.

(b) A certain stimulus administered to each of the 12 patients resulted the following increments of blood pressure: 6, 2, 8, -1, 8, 0, -2, 1, 5, 0, 5, 7. Can it be concluded that the stimulus has significant effect in increasing the blood pressure. (06)
(Take $t_{0.05} = 2.2$)

(c) Explain Sampling Theory with Examples. (06)

Q.3. (a) Find the root of the equation $\mathbf{x\log_{10}x - 1.2 = 0}$ by Regula falsi method correct upto three decimals. (06)

(b) Solve the following equations by Jacobi iterative method. (06)
$20x + y - 2z = 17;\ \ 3x + 20y - z = -18;\ \ 2x - 3y + 20z = 25.$

(c) Solve the differential equation $\frac{dy}{dx} = x + y^2$ by Runge's Kutta method with $y(0) = 1$ and find y at x = 0.2 in steps of 0.1. (06)

Q.4. (a) Find the polynomial $f(x)$ by using Langrange's formula and hence find $f(3)$: (06)

x	0	1	2	5
$f(x)$	2	3	12	147

(b) Find the value of $f(22)$ and $f(42)$ from the following available data : (06)

x	20	25	30	35	40	45
$f(x)$	354	332	291	260	231	204

(c) Evaluate $\int_0^1 \log\cos x\,dx$ using (a) Trapezoidal rule and (b) Simpson's one third rule. (06)

PTO

$x - x^2$

Q.5. (a) Find the Fourier series of the function $f(x) = -x^2$, for $-\pi$ to π. (06)

(b) Find the Fourier Transform of (06)

$$f(x) = 1 - x^2 , |x| \le 1$$
$$= 0, \qquad |x| \ge 1$$

Hence evaluate $\int_0^\infty \frac{x\cos x - \sin x}{x^3} \cos\frac{x}{2}\, dx$.

(c) Obtained the Fourier expansion of $f(x) = x \sin x$ as a cosine series in $(0, \pi)$. (06)

Hence show that $\frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \ldots\ldots\infty = \frac{\pi-2}{4}$.

***

ODE: 3V

RATION:

Total five

question c

clearly

2024050

B.E. II EXAMINATION DEC.' 2023
E&I
3AERC1: Applied Mathematics-III

Time: 3 Hrs.] **[Max. Marks: 60**

Note: Attempt any **TWO** parts from each Question. Each carries equal marks.

Q.1. (a) If $u - v = \dfrac{e^x - \cos x + \sin x}{\cosh y - \cos x}$ and $f(z) = u + iv$ is an analytic function, find $f(z)$ in terms of z by Milne Thomson method. (06)

(b) Using Cauchy's Integral formula, evaluate $\int_C \dfrac{\cos \pi z^2}{(z-1)(z-2)} dz$, where C is the circle $|z| = 3$. (06)

(c) Use residue calculus to evaluate the following integral $\int_0^{2\pi} \dfrac{\cos 3\theta}{5 - 4\cos\theta} d\theta$. (06)

Q.2. (a) A random variable X has the density function (06)

$$f(x) = \begin{cases} ax & 0 \le x < 1 \\ a & 1 \le x \le 2 \\ -ax + 3a & 2 < x \le 3 \\ 0 & \text{otherwise} \end{cases}$$

Determine 'a' and compute $P(X \le 1.5)$.

(b) Prove that a square matrix $A = \left[a_{ij}\right]$ has a fixed point if and only if the determinant of the matrix $(A - I)$ is zero i.e., $|A - I| = 0$, where I is unit matrix. (06)

(c) The time till failure of a part, T years has probability density function $f(t) = kt^{-4}$ $(t > 1)$ and is zero elsewhere, where k is a constant. (06)

1) Find the value of k
2) Find the mean time till failure
3) Find the failure rate

Q.3 (a) Obtain the missing term in the following table: (06)

x:	2.0	2.1	2.2	2.3	2.4	2.5	2.6
y:	0.135	------	0.111	0.100	-----	0.082	0.074

(b) Find the first & second derivatives of the function f(x) at point $x = 1.1$, where: (06)

x:	1.0	1.2	1.4	1.6	1.8	2.0
f(x):	0.00	0.1280	05440	1.2960	1.4320	4.000

(c) When a train is moving at 30m/sec steam is shut off and brakes are applied. The speed of the train per second after t sec is given by: - (06)

Time(t)	0	5	10	15	20	25	30	35	40
Speed(v)	30	24	19.5	16	13.6	11.7	10.0	8.5	7.5

Using Simpson's rule, determine the distance moved by the train in **40 secs**.

4 (a) Find the real root of the equation $x^3 - 3x + 1 = 0$ correct to four decimal places by using Newton-Raphson method (06)

PTO